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Task 4

Password Security Analysis

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Password Security Policy

1. Introduction

A password is used to protect user accounts and sensitive information from unauthorized access. A strong password policy helps an organization reduce the risk of password attacks and account compromise.

2. Password Requirements

- Password should be at least 12 characters long.
- Use uppercase letters, lowercase letters, numbers, and special characters.
- Use a unique password for every account.
- Do not use personal information such as name, birthday, phone number, or username.
- Avoid common passwords such as 123456, password, or qwerty.
- Never share passwords with others.
- Do not write passwords where unauthorized people can see them.
- Change the password immediately if it is compromised.

3. Password Uniqueness

Users should use different passwords for different accounts. If the same password is used on multiple websites and one website is hacked, attackers may use the leaked password to access other accounts.

Example: Using the same password for Gmail and an online banking account is unsafe.

4. Password Managers

A password manager securely stores and manages passwords. It can generate strong and unique passwords for different accounts, so users do not have to remember every password.

5. Multi-Factor Authentication (MFA)

MFA provides an additional layer of security by requiring more than one method of authentication.

Example: A user enters a password and then enters an OTP received on their mobile phone.

6. Password Security Best Practices

1. Create long and unique passwords.
2. Use a password manager.
3. Enable MFA whenever available.
4. Never share passwords.
5. Avoid using personal information in passwords.
6. Do not reuse passwords.
7. Be careful of phishing emails and fake login pages.
8. Change compromised passwords immediately.

7. Conclusion

Strong and unique passwords, password managers, and MFA help protect accounts from unauthorized access. Following a proper password security policy improves the overall security of an organization.

Interview Questions

1. What makes a password strong?

A strong password is **long, unique, and difficult to guess**. It should not contain easily available personal information. **Example:** River!Cloud#72Moon

2. Why should passwords be unique?

Passwords should be unique because if one account is hacked, attackers cannot use the same password to access other accounts.

3. What is MFA?

MFA (Multi-Factor Authentication) is a security method that requires two or more verification factors to access an account. **Example:** Password + OTP.

